EE656: Robust Control
Quiz 2

Model Answer

Set-1

`Marks will be awarded if and only if the correct steps leading to the answers are shown. You won't get marks for just writing the answer.`

`Hints: In order to disprove a sentence -- you can give a counter example to the sentence. In order to prove a statement -- you need to give the complete proof.`

`Disclaimer: Some questions` *might* `not have answers -- you need to justify the reason for no answer in such cases.`

Q1: Prove or disprove the following: $7 \times 2 = 14$

(a) The function space $\mathcal{L}_2$ and $\mathcal{L}_1$ satisfies the following inclusion $\mathcal{L}_2 \subset \mathcal{L}_1$.
Solution: False. See Example 3 and 4 in *Examples function space* file.

(b) If $f \in \mathcal{L}_2(0, \infty)$, then $f \in \mathcal{L}_2(-\infty, \infty)$.
Solution: True. Since $f \in \mathcal{L}_2(0, \infty)$, we have $\int_0^\infty (f^* f) dt < \infty \Rightarrow \int_{-\infty}^\infty (f^* f) dt < \infty$ (since $f = 0$ for $t \in (-\infty, 0]$). Therefore, $f \in \mathcal{L}_2(-\infty, \infty)$.

(c) If $v \in \mathcal{L}_\infty$, then there exists v_1 and v_2 such that $v = v_1 + v_2$ and $v_1 \in \mathcal{H}_\infty$ and $v_2 \in \mathcal{H}_\infty^\perp$.
Solution: False. There is no notion of orthogonality in $\mathcal{L}_\infty$ space, since $\mathcal{L}_\infty$ is a Banach space without the notion of inner product. Hence, the space $\mathcal{H}_\infty^\perp$ does not exists.

(d) $G(s) \in \mathcal{RH}_\infty$ if and only if $\texttt{det}(G(s))$ has no zeros in $\mathbb{C}^+$.
Solution: False. For example let $G(s) = \frac{s-1}{s+1}$. Clearly, $G(s) \in \mathcal{RH}_\infty$. However, $1 \in \mathbb{C}^+$ is a zero of $\texttt{det}(G(s))$.

(e) The feedback configuration in Figure 1 is internally stable if and only if the roots of $1 + PK$ are in $\mathbb{C}^-$ (Here r represents the reference signal).
Solution: False. If there are unstable pole zero cancellations of plant P and controller K, then the statement does not hold.

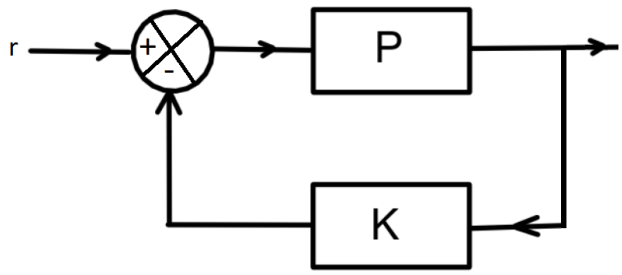

Figure 1: A feedback interconnection (Question 1e)

(f) The set of all power signals is a normed space.
Solution: False. The signal $f(t) = e^{-t}u(t)$ has zero power. This violates the norm condition $\|f\| = 0 \Leftrightarrow f = 0$.

(g) Let $\mathcal{F} : \ell_2 \to \mathcal{L}_2[-\pi, \pi]$ such that $\mathcal{F}(x) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$. Then, the operator $\mathcal{F} \in \mathcal{B}(\ell_2, \mathcal{L}_2[-\pi, \pi])$.
Solution: First you need to prove linearity, i.e., $\mathcal{F}(x_1 + x_2) = \mathcal{F}(x_1) + \mathcal{F}(x_2)$ (Skipping this part as it is easy to prove, you can prove this). Next we prove boundedness.

$$\begin{aligned}\|\mathcal{F}(x)\|_{\mathcal{L}_2}^2 &= \int_{-\pi}^{\pi} \Big| \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}\Big|^2 d\omega \\ &= \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} x[n]x^*[m] \int_{-\pi}^{\pi} e^{-j\omega(n-m)} d\omega = 2\pi \sum_{n=-\infty}^{\infty} |x[n]|^2 = 2\pi \|x\|_{\ell_2}^2.\end{aligned}$$

Since $\|\mathcal{F}\|_2 = \mathtt{sup}_{x\neq 0} \frac{\|\mathcal{F}(x)\|_{\mathcal{L}_2}}{\|x\|_{\ell_2}}$, we have $\|\mathcal{F}\|_2 = 2\pi$. Therefore, $\mathcal{F}$ is bounded.

Q2: You are control engineer entrusted with the task to estimate the effect of disturbance on a system with transfer function $P = \frac{1}{10s+1}$. After a detailed inspection you inferred that the disturbance input to the system have the energy bound $\|d\|_2 \leqslant 0.4$. What is the best estimate of the ∞-norm of the output?
Solution: $\|y\|_\infty \leqslant \|P\|_2 \|d\|_2 \Rightarrow \frac{1}{\sqrt{20}} \times 0.4 \Rightarrow \|y\|_\infty \leqslant 0.0894$.
Calculation of $\|P\|_2$: A minimal state-space realization of P is $(-\frac{1}{10}, \frac{1}{10}, 1, 0)$.
The Lyapunov equation is: $AW_c + W_cA^T + BB^T = 0 \Rightarrow -\frac{2}{10}W_c + \frac{1}{100} = 0 \Rightarrow W_c = \frac{1}{20}$.
Therefore, $\|P\|_2 = \sqrt{CW_cC^T} = \frac{1}{\sqrt{20}}$.

Q3: Find the $\mathcal{L}_\infty$ norm of G_1, G_2, and G_3 with the information given below: $3 \times 5 = 15$

(a) $G_1(s)$ admits the Bode-magnitude plot in Figure 2.
Solution: $\|G_1(s)\|_\infty = \mathtt{ess\,sup}_\omega\{\sigma_{\mathtt{max}}[G(j\omega)]\}$. Hence for SISO case, it is the maximum gain attainable. From the plot, the maximum magnitude is 19.1 dB. Hence,

$$20\log(\|G_1(s)\|_\infty) = 19.1 \Rightarrow \|G_1(s)\|_\infty = 2.599.$$

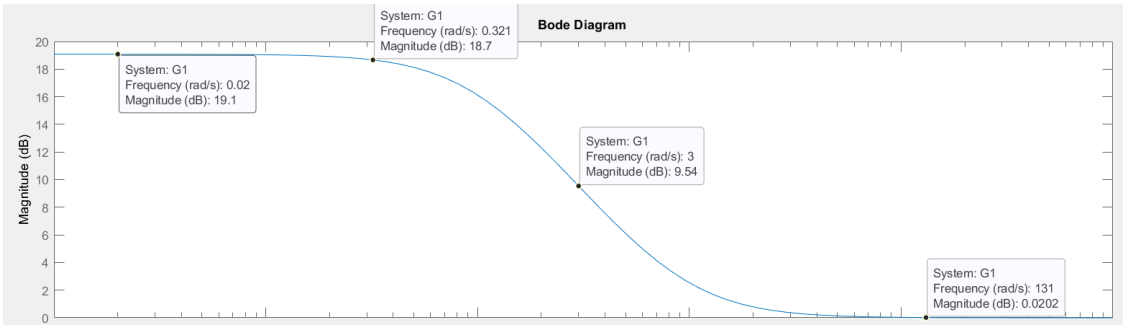

Figure 2: Bode plot of $G_1(s)$ (Question 3a)

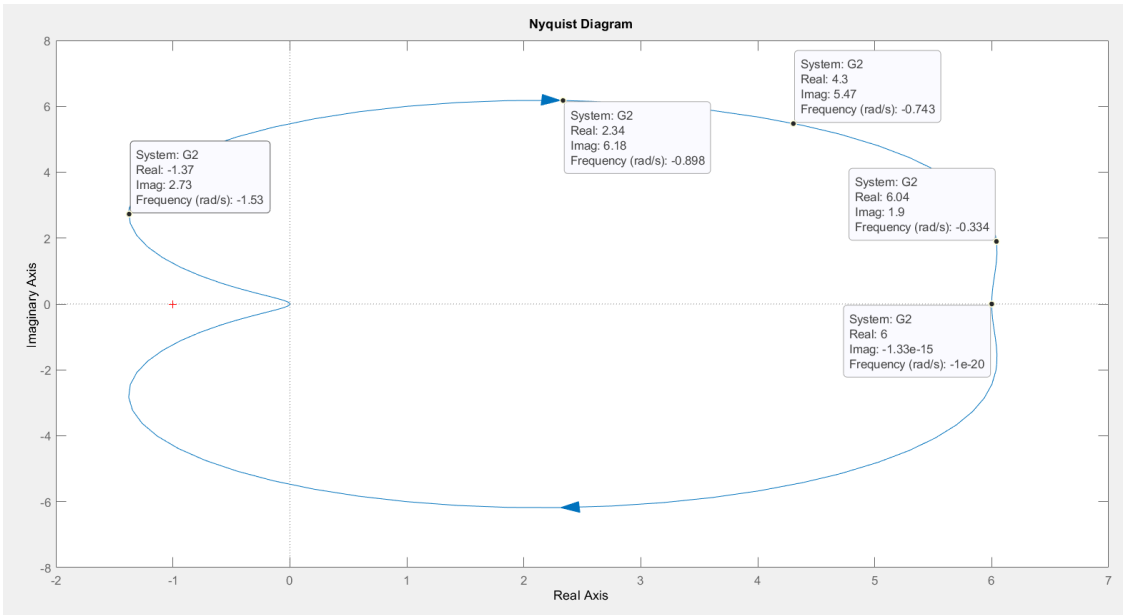

Figure 3: Nyquist plot of $G_2(s)$ (Question 3b)

(b) $G_2(s)$ has the following Nyquist plot (Figure 3).

Solution: From the plot, the approximate $\mathcal{L}_\infty$ norm is: $\|G_1(s)\|_\infty \approx \sqrt{4.3^2 + 5.47^2} = 6.958$.

(c) $G_3(s) = \frac{s^2-4}{s^2+s-6}$.

Solution: $G_3(s) = \frac{s^2-4}{s^2+s-6} = \frac{(s+2)}{(s+3)}$. Two ways to do it - (i) You can find the Bode/Nyquist plot and find. (ii) Find the values of ω where the maximum is attained. We show this method here.

$$\frac{d}{d\omega}|G_3(j\omega)|^2 = \frac{d}{d\omega}\left(\frac{4+\omega^2}{9+\omega^2}\right) = 0 \Rightarrow \frac{2\omega(13+2\omega^2)}{(9+\omega^2)^2} = 0 \Rightarrow \omega = 0, \infty.$$

The maxima is attained at $\omega = \infty$. Thus we have $\lim_{\omega\to\infty} |G_3(j\omega)|^2 = 1$. Therefore, $\|G_3(s)\|_\infty = 1$.

Q4: Define the following transfer matrices: $10 + 16 = 26$

$$P_1(s) = \begin{bmatrix} \frac{1}{3+4s+s^2} & \frac{-1}{6+11s+6s^2+s^3} & 0 \\ \frac{-1}{6+11s+6s^2+s^3} & \frac{1}{3+4s+s^2} & 0 \\ \frac{-1}{6+11s+6s^2+s^3} & \frac{1}{12+28s+23s^2+8s^3+s^4} & \frac{1}{4+4s+s^2} \end{bmatrix},$$

$$P_2(s) = \begin{bmatrix} \frac{-1+4s+2s^2}{-6-5s+2s^2+s^3} & \frac{-1}{3+4s+s^2} & 0 \\ \frac{-1}{3+4s+s^2} & \frac{-5+2s+2s^2}{-12-13s+s^3} & 0 \\ \frac{-1}{3+4s+s^2} & \frac{1}{6+11s+6s^2+s^3} & \frac{1+2s}{-2+s+s^2} \end{bmatrix}$$

(a) Find a minimal realization of $P_1(s)$ and $P_2(s)$. (You are not allowed to use inbuilt functions in software packages).

(b) Find the $\mathcal{L}_2$-norm and $\mathcal{H}_2$-norm of $P_1(s)$ and $P_2(s)$. (You are not allowed to use inbuilt functions to directly compute the norms).

Solution: (a) Partial fraction of $P_1(s)$ gives

$$P_1(s) = \begin{bmatrix} \frac{1}{(s+1)(s+3)} & \frac{-1}{(s+1)(s+2)(s+3)} & 0 \\ \frac{-1}{(s+1)(s+2)(s+3)} & \frac{1}{(s+1)(s+3)} & 0 \\ \frac{-1}{(s+1)(s+2)(s+3)} & \frac{1}{(s+1)(s+2)^2(s+3)} & \frac{1}{(s+2)^2} \end{bmatrix} \tag{1}$$

$$= \underbrace{\begin{bmatrix} \frac{(s+2)}{(s+1)(s+3)} & \frac{-1}{(s+1)(s+3)} & 0 \\ \frac{-1}{(s+1)(s+3)} & \frac{(s+2)}{(s+1)(s+3)} & 0 \\ \frac{-1}{(s+1)(s+3)} & \frac{1}{(s+1)(s+2)(s+3)} & \frac{1}{(s+2)} \end{bmatrix}}_{P_{11}(s))} \underbrace{\begin{bmatrix} \frac{1}{s+2} & 0 & 0 \\ 0 & \frac{1}{s+2} & 0 \\ 0 & 0 & \frac{1}{s+2} \end{bmatrix}}_{P_{12}(s)} \tag{2}$$

We find the minimal representation $(A_{11}, B_{11}, C_{11}, D_{11})$ for P_{11} first. Partial fraction of $P_{11}(s)$ gives

$$\begin{bmatrix} \frac{(s+2)}{(s+1)(s+3)} & \frac{-1}{(s+1)(s+3)} & 0 \\ \frac{-1}{(s+1)(s+3)} & \frac{(s+2)}{(s+1)(s+3)} & 0 \\ \frac{-1}{(s+1)(s+3)} & \frac{1}{(s+1)(s+2)(s+3)} & \frac{1}{(s+2)} \end{bmatrix} = \frac{1}{s+1}\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} + \frac{1}{s+2}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -1 & 1 \end{bmatrix} + \frac{1}{s+3}\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

Therefore, by Gilbert's method, we have

$$A_{11} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}, B_{11} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & -1 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}, C_{11} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 1 & 1 \end{bmatrix}, D_{11} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

We find the minimal representation $(A_{21}, B_{22}, C_{22}, D_{22})$ for P_{12} next.

$$A_{12} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}, B_{12} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, C_{12} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, D_{12} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Therefore, a state-space representation (A_1, B_1, C_1, D_1) of $P_1(s)$ is

$$P_1(s) = \left[\begin{array}{c|c} A_1 & B_1 \\ C_1 & D_1 \end{array}\right] = \left[\begin{array}{cc|c} A_{11} & B_{11}C_{12} & B_{11}D_{12} \\ 0 & A_{12} & B_{12} \\ \hline C_{11} & D_{11}C_{12} & D_{11}D_{12} \end{array}\right], \text{ where}$$

$$A_1 = \begin{bmatrix} -1 & 0 & 0 & 0.5 & -0.5 & 0 \\ 0 & -2 & 0 & 0 & -1 & 1 \\ 0 & 0 & -3 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & -2 \end{bmatrix}, B_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$C_1 = \frac{1}{2}\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 \\ -1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}, D_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

It is easy to check that (A_1, B_1, C_1) is controllable & observable and hence, minimal.

Solve the Lyapunov equation $AW_c + W_cA^T + BB^T = 0$ and then compute $\texttt{Trace}(BW_cB^T) = \|P_1(s)\|_2^2 = 0.1413$. Therefore, $\|P_1(s)\|_{\mathcal{H}_2} = \|P_1(s)\|_{\mathcal{L}_2} = 0.3760$.

(b) Partial fraction of $P_2(s)$ gives

$$P_2(s) = P_{11}(s) + H_1(s), \text{where } H_1(s) = \begin{bmatrix} \frac{1}{s-2} & 0 & 0 \\ 0 & \frac{1}{s-4} & 0 \\ 0 & 0 & \frac{1}{s-1} \end{bmatrix}$$

Clearly, $P_2(s) \notin \mathcal{H}_2$ and we cannot compute the $\mathcal{H}_2$ norm. However, we can compute the $\mathcal{L}_2$ norm, since $P_2(s) \in \mathcal{RL}_2$. Since $\|P_2(s)\|_{\mathcal{L}_2}^2 = \|P_{11}(s)\|_{\mathcal{H}_2}^2 + \|H_1(-s)\|_{\mathcal{H}_2}^2$, we calculate $\|H_1(-s)\|_{\mathcal{H}_2}$ next.

Minimal representation of $H_1(-s) = \begin{bmatrix} \frac{-1}{s+1} & 0 & 0 \\ 0 & \frac{-1}{s+4} & 0 \\ 0 & 0 & \frac{-1}{s+1} \end{bmatrix}$ is

$$H_1(-s) = \left[\begin{array}{ccc|ccc} -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -4 & 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 0 & 0 & 1 \\ \hline -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{array}\right]$$

Now solving the Lyapunov equation and using the trace equation, one can calculate the $\|H_1(-s)\|_{\mathcal{H}_2} = 0.9354$.

Next we find $\|P_{11}(s)\|_{\mathcal{H}_2}$. The minimal representation of $P_{11}(s)$ is given by $(A_{11}, B_{11}, C_{11}, D_{11})$. Again solving the Lyapunov equation and using trace equation will give $\|P_1(s)\|_{\mathcal{H}_2} = 0.9832$. Therefore, $\|P_2(s)\|_{\mathcal{L}_2} = \sqrt{0.9832^2 + 0.9354^2} = 1.357$.

EE656: Robust Control
Quiz 2

Set-2

Q1: Prove or disprove the following: $7 \times 2 = 14$

(a) The function space $\mathcal{L}_1$ and $\mathcal{L}_2$ satisfies the following inclusion $\mathcal{L}_1 \subset \mathcal{L}_2$.
Solution: False. See Example 3 and 4 in *Examples function space* file.

(b) If $f \in \mathcal{L}_2[a, b]$, then $f \in \mathcal{L}_2(-\infty, \infty)$, where $a, b \in \mathbb{R}$.
Solution: True. Since $f \in \mathcal{L}_2(a, b)$, we have $\int_a^b (f^* f)dt < \infty \Rightarrow \int_{-\infty}^{\infty} (f^* f)dt < \infty$ (since $f = 0$ for $t \in (-\infty, a] \cup [b, \infty)$). Therefore, $f \in \mathcal{L}_2(-\infty, \infty)$.

(c) If $v \in \mathcal{L}_\infty$, then there exists v_1 and v_2 such that $v = v_1 + v_2$ and $v_1 \in \mathcal{H}_\infty$ and $v_2 \in \mathcal{H}_\infty^\perp$.
Solution: False. There is no notion of orthogonality in $\mathcal{L}_\infty$ space, since $\mathcal{L}_\infty$ is a Banach space without the notion of inner product. Hence, the space $\mathcal{H}_\infty^\perp$ does not exists.

(d) $G(s) \in \mathcal{RH}_\infty$ if $\texttt{det}(G(s))$ has no zeros in $\mathbb{C}^+$.
Solution: False. For example let $G(s) = \frac{s+1}{s-1}$. Clearly, $G(s) \notin \mathcal{RH}_\infty$ although there are no zeros in $\mathbb{C}^+$.

(e) The feedback configuration in Figure 1 is internally stable if and only if the roots of $1 + PK$ are in $\mathbb{C}^-$ (Here r represents the reference signal).

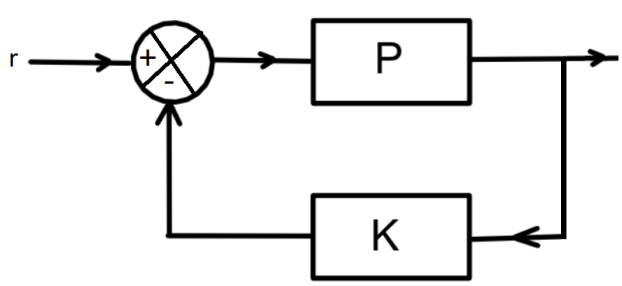

Figure 1: A feedback interconnection (Question 1e)

Solution: False. If there are unstable pole zero cancellations of plant P and controller K, then the statement does not hold.

(f) The set of all power signals is a normed space.
Solution: False. The signal $f(t) = e^{-t}u(t)$ has zero power. This violates the norm condition $\|f\| = 0 \Leftrightarrow f = 0$.

(g) Let $\mathcal{F} : \ell_2 \to \mathcal{L}_2[-\pi, \pi]$ such that $\mathcal{F}(x) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$. Then, $\|\mathcal{F}\|_2 = \sqrt{2}\pi$.
Solution: First you need to prove linearity, i.e., $\mathcal{F}(x_1 + x_2) = \mathcal{F}(x_1) + \mathcal{F}(x_2)$ (Skipping this part as it is easy to prove, you can prove this). Next we prove boundedness.

$$\begin{aligned}\|\mathcal{F}(x)\|_{\mathcal{L}_2}^2 &= \int_{-\pi}^{\pi} \Big| \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \Big|^2 d\omega \\ &= \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} x[n]x^*[m] \int_{-\pi}^{\pi} e^{-j\omega(n-m)} d\omega = 2\pi \sum_{n=-\infty}^{\infty} |x[n]|^2 = 2\pi \|x\|_{\ell_2}^2.\end{aligned}$$

Since $\|\mathcal{F}\|_2 = \mathtt{sup}_{x\neq 0} \frac{\|\mathcal{F}(x)\|_{\mathcal{L}_2}}{\|x\|_{\ell_2}}$, we have $\|\mathcal{F}\|_2 = 2\pi$. Therefore, $\mathcal{F}$ is bounded.

Q2: You are control engineer entrusted with the task to estimate the effect of disturbance on a system with transfer function $P = \frac{1}{15s+2}$. After a detailed inspection you infered that the disturbance input to the system have the energy bound $\|d\|_2 \leqslant 0.6$. What is the best estimate of the ∞-norm of the output?
Solution: $\|y\|_\infty \leqslant \|P\|_2 \|d\|_2 \Rightarrow \frac{1}{\sqrt{60}} \times 0.6 \Rightarrow \|y\|_\infty \leqslant 0.077$.
Calculation of $\|P\|_2$: A minimal state-space realization of P is $(-\frac{2}{15}, \frac{1}{15}, 1, 0)$.
The Lyapunov equation is: $AW_c + W_cA^T + BB^T = 0 \Rightarrow -\frac{4}{15}W_c + \frac{1}{125} = 0 \Rightarrow W_c = \frac{1}{60}$.
Therefore, $\|P\|_2 = \sqrt{CW_cC^T} = \frac{1}{\sqrt{60}}$.

Q3: Find the $\mathcal{L}_\infty$ norm of G_1, G_2, and G_3 with the information given below: $3 \times 5 = 15$

(a) $G_1(s)$ admits the Bode-magnitude plot in Figure 2.

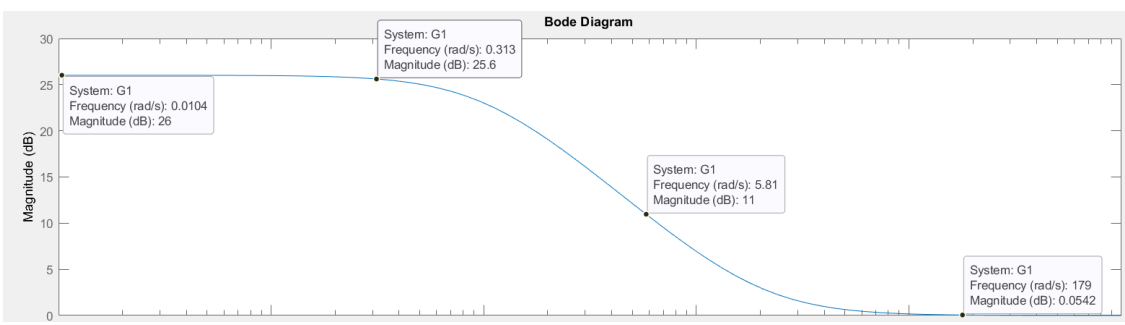

Figure 2: Bode plot of $G_1(s)$ (Question 3a)

Solution: $\|G_1(s)\|_\infty = \mathtt{ess\,sup}_\omega\{\sigma_{\mathtt{max}}[G(j\omega)]\}$. Hence for SISO case, it is the maximum gain attainable. From the plot, the maximum magnitude is 26 dB. Hence,

$$20\log(\|G_1(s)\|_\infty) = 26 \Rightarrow \|G_1(s)\|_\infty = 3.67.$$

(b) $G_2(s)$ has the following Nyquist plot (Figure 3).
Solution: From the plot, the approximate $\mathcal{L}_\infty$ norm is: $\|G_1(s)\|_\infty \approx \sqrt{16^2 + 16.4^2} = 22.912$.

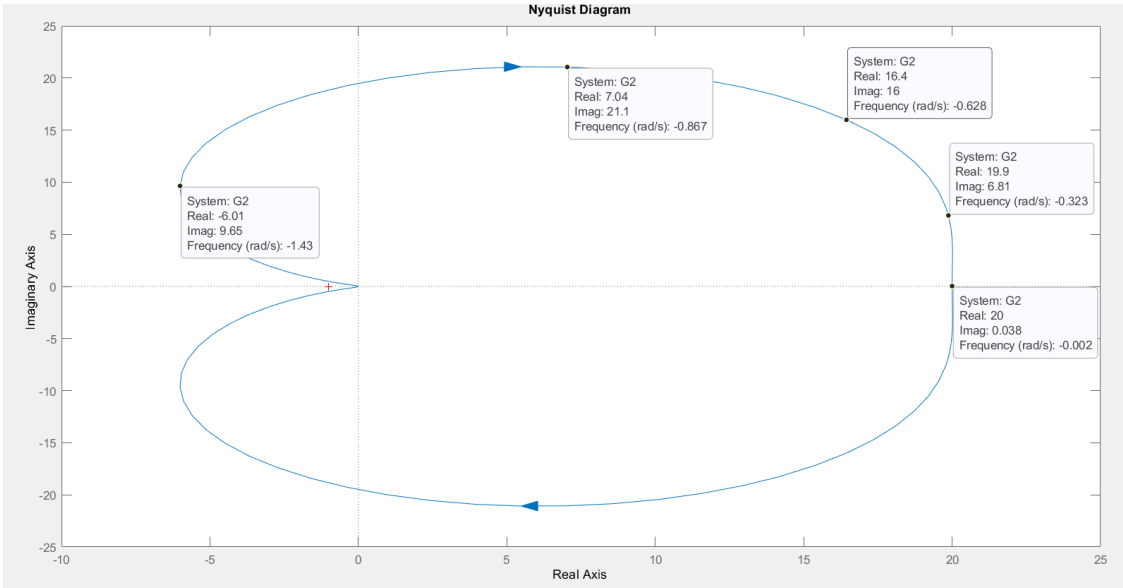

Figure 3: Nyquist plot of $G_2(s)$ (Question 3b)

(c) $G_3(s) = \frac{s^2+2s+1}{s^2+2s-3}$.
Solution: $G_3(s) = \frac{s^2+2s+1}{s^2+2s-3} = \frac{(s+1)^2}{(s+3)(s-1)}$. Two ways to do it - (i) You can find the Bode/Nyquist plot and find. (ii) Find the values of ω where the maximum is attained. We show this method here.

$$\frac{d}{d\omega}|G_3(j\omega)|^2 = \frac{d}{d\omega}\left(\frac{1+\omega^2}{9+\omega^2}\right) = 0 \Rightarrow \frac{2\omega(13+2\omega^2)}{(9+\omega^2)^2} = 0 \Rightarrow \omega = 0, \infty.$$

The maxima is attained at $\omega = \infty$. Thus we have $\lim_{\omega\to\infty} |G_3(j\omega)|^2 = 1$. Therefore, $\|G_3(s)\|_\infty = 1$.

Q4: Define the following transfer matrices: $10 + 16 = 26$

$$P_1(s) = \begin{bmatrix} \frac{2+s}{3+7s+5s^2+s^3} & \frac{-1}{3+7s+5s^2+s^3} & 0 \\ \frac{-1}{3+7s+5s^2+s^3} & \frac{2+s}{3+7s+5s^2+s^3} & 0 \\ \frac{-1}{3+7s+5s^2+s^3} & \frac{1}{6+17s+17s^2+7s^3+s^4} & \frac{1}{2+3s+s^2} \end{bmatrix},$$

$$P_2(s) = \begin{bmatrix} \frac{-3+3s+2s^2}{-9-9s+s^2+s^3} & \frac{-1}{3+4s+s^2} & 0 \\ \frac{-1}{3+4s+s^2} & \frac{-1+4s+2s^2}{-6-5s+2s^2+s^3} & 0 \\ \frac{-1}{3+4s+s^2} & \frac{1}{6+11s+6s^2+s^3} & \frac{1+2s}{-2+s+s^2} \end{bmatrix}$$

(a) Find a minimal realization of $P_1(s)$ and $P_2(s)$. (You are not allowed to use inbuilt functions in software packages).

(b) Find the $\mathcal{L}_2$-norm and $\mathcal{H}_2$-norm of $P_1(s)$ and $P_2(s)$. (You are not allowed to use inbuilt functions to directly compute the norms).

Solution: Check the solution of Part-A for details of the procedure. Here

$$P_1(s) = \underbrace{\begin{bmatrix} \frac{s+2}{s^2+4s+3} & \frac{-1}{s^2+4s+3} & 0 \\ \frac{-1}{s^2+4s+3} & \frac{s+2}{s^2+4s+3} & 0 \\ \frac{-1}{s^2+4s+3} & \frac{1}{s^3+6s^2+11s+6} & \frac{1}{s+2} \end{bmatrix}}_{P_{11}(s)} \underbrace{\begin{bmatrix} \frac{1}{s+1} & 0 & 0 \\ 0 & \frac{1}{s+1} & 0 \\ 0 & 0 & \frac{1}{s+1} \end{bmatrix}}_{P_{12}(s)}$$

Thus,

$$A_{11} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}, B_{11} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & 1 \\ 1 & 1 & 0 \end{bmatrix} C_{11} = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ -\frac{1}{2} & 0 & \frac{1}{2} \\ -\frac{1}{2} & 1 & \frac{1}{2} \end{bmatrix}, D_{11} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$A_{12} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, B_{12} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, C_{12} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, D_{12} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

A minimal representation is:

$$A = \begin{bmatrix} -1 & 0 & 0 & 1 & -1 & 0 \\ 0 & -2 & 0 & 0 & -1 & 1 \\ 0 & 0 & -3 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix}, B_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$C_1 = \frac{1}{2} \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 \\ -1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}, D_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Following the steps in Part-A, we will get $\|P_1\|_{\mathcal{H}_2} = 0.6379$.

Check that

$$P_2(s) = P_{11}(s) + H_1(s), \text{ where } H_1(s) = \begin{bmatrix} \frac{1}{s-3} & 0 & 0 \\ 0 & \frac{1}{s-2} & 0 \\ 0 & 0 & \frac{1}{s-1} \end{bmatrix}$$

Clearly, $P_2(s) \notin \mathcal{H}_2$ and we cannot compute the $\mathcal{H}_2$ norm. However, we can compute the $\mathcal{L}_2$ norm, since $P_2(s) \in \mathcal{RL}_2$. Here, $\|P_2(s)\|^2_{\mathcal{L}_2} = \|P_{11}(s)\|^2_{\mathcal{H}_2} + \|H_1(-s)\|^2_{\mathcal{H}_2} = 0.9831^2 + 0.9574^2 \Rightarrow \|P_2(s)\|_{\mathcal{L}_2} = 1.3723$.

EE656: Robust Control
Quiz 2

Set-3

Q1: Prove or disprove the following: $7 \times 2 = 14$

(a) The function space $\mathcal{L}_\infty$ and $\mathcal{L}_2$ satisfies the following inclusion $\mathcal{L}_\infty \subset \mathcal{L}_2$.
Solution: False. See Example 9 in *Examples function space* file.

(b) If $f \in \mathcal{L}_2[-\infty, 6)$, then $f \in \mathcal{L}_2(-\infty, \infty)$.
Solution: True. Since $f \in \mathcal{L}_2(a, b)$, we have $\int_{-\infty}^{6}(f^* f)dt < \infty \Rightarrow \int_{-\infty}^{\infty}(f^* f)dt < \infty$ (since $f = 0$ for $t \in [6, \infty)$). Therefore, $f \in \mathcal{L}_2(-\infty, \infty)$.

(c) If $v \in \mathcal{L}_\infty$, then there exists v_1 and v_2 such that $v = v_1 + v_2$ and $v_1 \in \mathcal{H}_\infty$ and $v_2 \in \mathcal{H}_\infty^\perp$.
Solution: False. There is no notion of orthogonality in $\mathcal{L}_\infty$ space, since $\mathcal{L}_\infty$ is a Banach space without the notion of inner product. Hence, the space $\mathcal{H}_\infty^\perp$ does not exists.

(d) If $G(s) \in \mathcal{RH}_\infty$, then $\texttt{det}(G(s))$ has no zeros in $\mathbb{C}^+$.
Solution: False. For example let $G(s) = \frac{s-1}{s+1}$. Clearly, $G(s) \in \mathcal{RH}_\infty$ although there are zeros in $\mathbb{C}^+$.

(e) The feedback configuration in Figure 1 is internally stable if and only if the roots of $1 + PK$ are in $\mathbb{C}^-$ (Here r represents the reference signal).

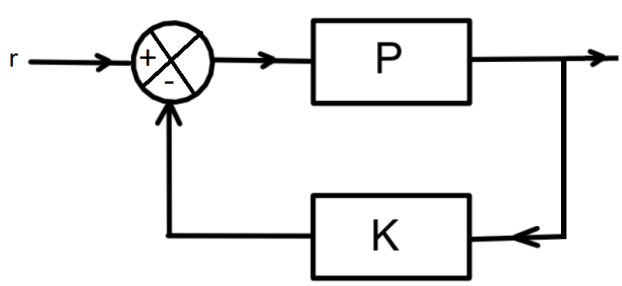

Figure 1: A feedback interconnection (Question 1e)

Solution: False. If there are unstable pole zero cancellations of plant P and controller K, then the statement does not hold.

(f) The set of all power signals is a normed space.
Solution: False. The signal $f(t) = e^{-t}u(t)$ has zero power. This violates the norm condition $\|f\| = 0 \Leftrightarrow f = 0$.

(g) Let $\mathcal{F} : \ell_2 \to \mathcal{L}_2[-\pi, \pi]$ such that $\mathcal{F}(x) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$. Then, the operator $\mathcal{F} \notin \mathcal{B}(\ell_2, \mathcal{L}_2[-\pi, \pi])$.

Solution: First you need to prove linearity, i.e., $\mathcal{F}(x_1 + x_2) = \mathcal{F}(x_1) + \mathcal{F}(x_2)$ (Skipping this part as it is easy to prove, you can prove this). Next we prove boundedness.

$$\begin{aligned}
\|\mathcal{F}(x)\|_{\mathcal{L}_2}^2 &= \int_{-\pi}^{\pi} \Big| \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \Big|^2 d\omega \\
&= \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} x[n]x^*[m] \int_{-\pi}^{\pi} e^{-j\omega(n-m)} d\omega = 2\pi \sum_{n=-\infty}^{\infty} |x[n]|^2 = 2\pi \|x\|_{\ell_2}^2.
\end{aligned}$$

Since $\|\mathcal{F}\|_2 = \mathtt{sup}_{x \neq 0} \frac{\|\mathcal{F}(x)\|_{\mathcal{L}_2}}{\|x\|_{\ell_2}}$, we have $\|\mathcal{F}\|_2 = 2\pi$. Therefore, $\mathcal{F}$ is bounded.

Q2: You are control engineer entrusted with the task to estimate the effect of disturbance on a system with transfer function $P = \frac{1}{6s+2}$. After a detailed inspection you inferred that the disturbance input to the system have the energy bound $\|d\|_2 \leqslant 0.1$. What is the best estimate of the ∞-norm of the output?

Solution: $\|y\|_\infty \leqslant \|P\|_2 \|d\|_2 \Rightarrow \frac{1}{\sqrt{24}} \times 0.1 \Rightarrow \|y\|_\infty \leqslant 0.0204$.

Calculation of $\|P\|_2$: A minimal state-space realization of P is $(-\frac{1}{3}, \frac{1}{6}, 1, 0)$.

The Lyapunov equation is: $AW_c + W_cA^T + BB^T = 0 \Rightarrow -\frac{2}{3}W_c + \frac{1}{36} = 0 \Rightarrow W_c = \frac{1}{24}$.

Therefore, $\|P\|_2 = \sqrt{CW_cC^T} = \frac{1}{\sqrt{24}}$.

Q3: Find the $\mathcal{L}_\infty$ norm of G_1, G_2, and G_3 with the information given below: $3 \times 5 = 15$

(a) $G_1(s)$ admits the Bode-magnitude plot in Figure 2.

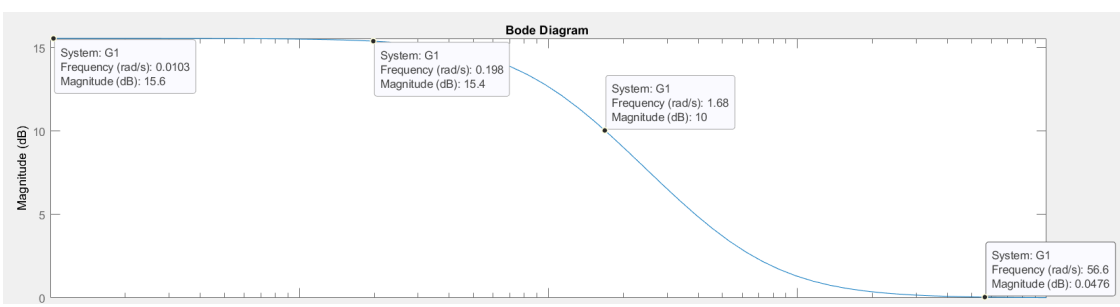

Figure 2: Bode plot of $G_1(s)$ (Question 3a)

Solution: $\|G_1(s)\|_\infty = \mathtt{ess\,sup}_\omega\{\sigma_{\mathtt{max}}[G(j\omega)]\}$. Hence for SISO case, it is the maximum gain attainable. From the plot, the maximum magnitude is 15.6 dB. Hence,

$$20\log(\|G_1(s)\|_\infty) = 15.6 \Rightarrow \|G_1(s)\|_\infty = 6.03.$$

(b) $G_2(s)$ has the following Nyquist plot (Figure 3).

Solution: From the plot, the approximate $\mathcal{L}_\infty$ norm is: $\|G_1(s)\|_\infty \approx \sqrt{8.01^2 + 6.67^2} = 10.423$.

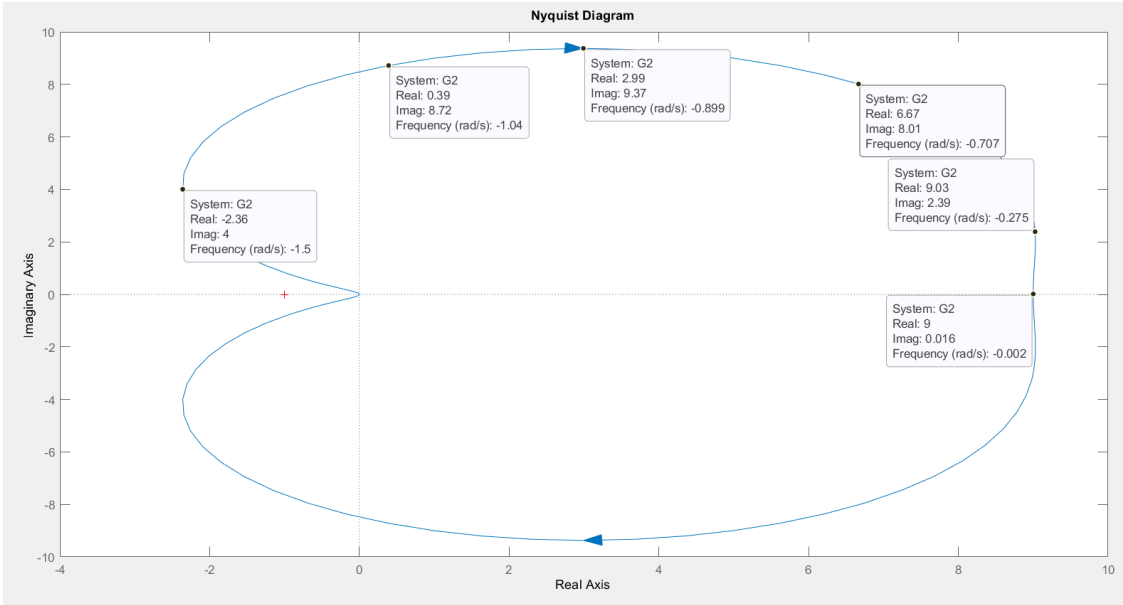

Figure 3: Nyquist plot of $G_2(s)$ (Question 3b)

(c) $G_3(s) = \frac{s^2+3s+2}{s^2-4}$.

Solution: $G_3(s) = \frac{s^2+3s+2}{s^2-4} = \frac{(s+1)}{(s-2)}$. Two ways to do it - (i) You can find the Bode/Nyquist plot and find. (ii) Find the values of ω where the maximum is attained. We show this method here.

$$\frac{d}{d\omega}|G_3(j\omega)|^2 = \frac{d}{d\omega}\left(\frac{1+\omega^2}{4+\omega^2}\right) = 0 \Rightarrow \frac{2\omega(5+2\omega^2)}{(4+\omega^2)^2} = 0 \Rightarrow \omega = 0, \infty.$$

The maxima is attained at $\omega = \infty$. Thus we have $\lim_{\omega\to\infty} |G_3(j\omega)|^2 = 1$. Therefore, $\|G_3(s)\|_\infty = 1$.

Q4: Define the following transfer matrices: $10 + 16 = 26$

$$P_1(s) = \begin{bmatrix} \frac{2+s}{9+15s+7s^2+s^3} & \frac{-1}{9+15s+7s^2+s^3} & 0 \\ \frac{-1}{9+15s+7s^2+s^3} & \frac{2+s}{9+15s+7s^2+s^3} & 0 \\ \frac{-1}{9+15s+7s^2+s^3} & \frac{1}{18+39s+29s^2+9s^3+s^4} & \frac{1}{6+5s+s^2} \end{bmatrix},$$

$$P_2(s) = \begin{bmatrix} \frac{-5+2s+2s^2}{-12-13s+s^3} & \frac{-1}{3+4s+s^2} & 0 \\ \frac{-1}{3+4s+s^2} & \frac{-1+4s+2s^2}{-6-5s+2s^2+s^3} & 0 \\ \frac{-1}{3+4s+s^2} & \frac{1}{6+11s+6s^2+s^3} & \frac{-1+2s}{-6-s+s^2} \end{bmatrix}$$

(a) Find a minimal realization of $P_1(s)$ and $P_2(s)$. (You are not allowed to use inbuilt functions in software packages).

(b) Find the $\mathcal{L}_2$-norm and $\mathcal{H}_2$-norm of $P_1(s)$ and $P_2(s)$. (You are not allowed to use inbuilt functions to directly compute the norms).

Solution: Check the solution of Part-A for details of the procedure. Here

$$P_1(s) = \underbrace{\begin{bmatrix} \frac{s+2}{s^2+4s+3} & \frac{-1}{s^2+4s+3} & 0 \\ \frac{-1}{s^2+4s+3} & \frac{s+2}{s^2+4s+3} & 0 \\ \frac{-1}{s^2+4s+3} & \frac{1}{s^3+6s^2+11s+6} & \frac{1}{s+2} \end{bmatrix}}_{P_{11}(s)} \underbrace{\begin{bmatrix} \frac{1}{s+3} & 0 & 0 \\ 0 & \frac{1}{s+3} & 0 \\ 0 & 0 & \frac{1}{s+3} \end{bmatrix}}_{P_{12}(s)}$$

Thus,

$$A_{11} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}, B_{11} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & 1 \\ 1 & 1 & 0 \end{bmatrix} C_{11} = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ -\frac{1}{2} & 0 & \frac{1}{2} \\ -\frac{1}{2} & 1 & \frac{1}{2} \end{bmatrix}, D_{11} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$A_{12} = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix}, B_{12} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, C_{12} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, D_{12} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

A minimal representation is:

$$A = \begin{bmatrix} -1 & 0 & 0 & 1 & -1 & 0 \\ 0 & -2 & 0 & 0 & -1 & 1 \\ 0 & 0 & -3 & 1 & 1 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 & -3 \end{bmatrix}, B_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$C_1 = \frac{1}{2} \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 \\ -1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}, D_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Following the steps in Part-A, we will get $\|P_1\|_{\mathcal{H}_2} = 0.2693$.

Check that

$$P_2(s) = P_{11}(s) + H_1(s), \text{ where } H_1(s) = \begin{bmatrix} \frac{1}{s-4} & 0 & 0 \\ 0 & \frac{1}{s-2} & 0 \\ 0 & 0 & \frac{1}{s-3} \end{bmatrix}$$

Clearly, $P_2(s) \notin \mathcal{H}_2$ and we cannot compute the $\mathcal{H}_2$ norm. However, we can compute the $\mathcal{L}_2$ norm, since $P_2(s) \in \mathcal{RL}_2$. Here, $\|P_2(s)\|^2_{\mathcal{L}_2} = \|P_{11}(s)\|^2_{\mathcal{H}_2} + \|H_1(-s)\|^2_{\mathcal{H}_2} = 0.9831^2 + 0.736^2 \Rightarrow \|P_2(s)\|_{\mathcal{L}_2} = 1.2281$.